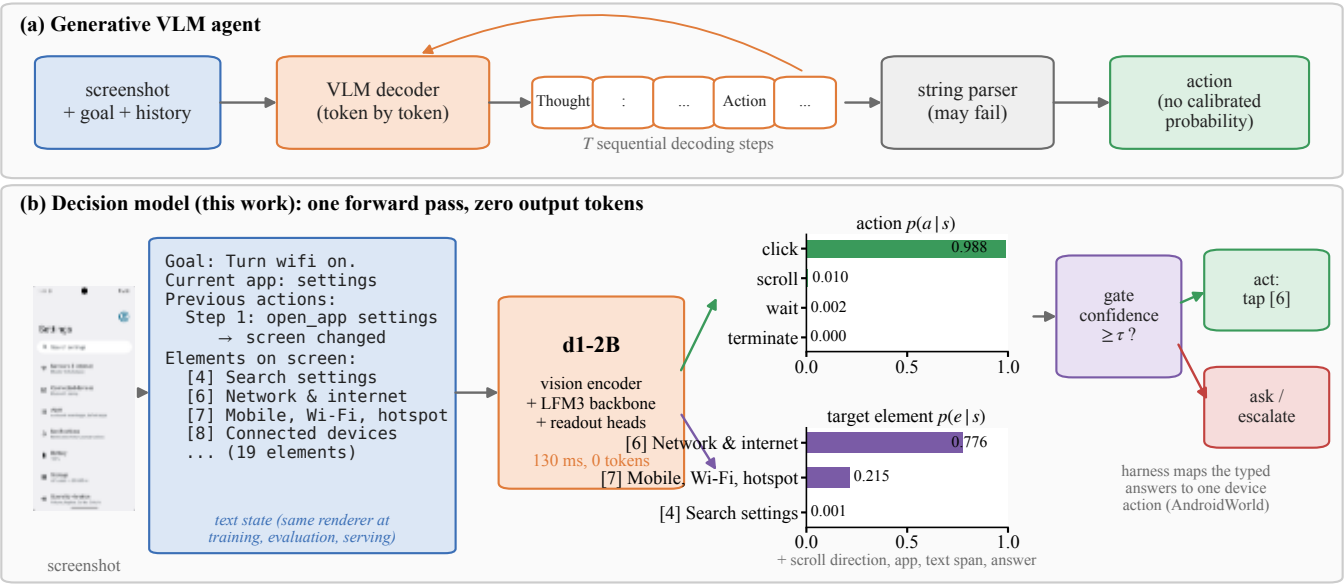

(a) Generative VLM agent
screenshot + goal + history
VLM decoder (token by token)
Thought
:
...
Action
...
T sequential decoding steps
string parser (may fail)
action (no calibrated probability)
(b) Decision model (this work): one forward pass, zero output tokens
Goal: Turn wifi on.
Current app: settings
Previous actions:
Step 1: open_app settings
→ screen changed
Elements on screen:
[4] Search settings
[6] Network & internet
[7] Mobile, Wi-Fi, hotspot
[8] Connected devices
... (19 elements)
text state (same renderer at training, evaluation, serving)
screenshot
d1-2B
vision encoder
+ LFM3 backbone
+ readout heads
130 ms, 0 tokens
action p(a | s)
click
0.988
scroll
0.010
wait
0.002
terminate
0.000
0.0
0.5
1.0
target element p(e | s)
[6] Network & internet
0.776
[7] Mobile, Wi-Fi, hotspot
0.215
[4] Search settings
0.001
0.0
0.5
1.0
+ scroll direction, app, text span, answer
gate confidence ≥ τ ?
act: tap [6]
ask / escalate
harness maps the typed answers to one device action (AndroidWorld)